

**FALCON
AUTOTECH**



EXPRESS PARCELS  INTERNATIONAL  E-COMMERCE

FALCON's Response to DTDC for Supply of LINEAR ARM Sorter System – 6000 PPH Including secondary sort via PTL

August 20, 2025



www.falconautotech.com

Kind Attention –

Mr. Ajinkya Kulkarni

DTDC

Offer Ref: F25-00170-05

Date 20-08-2025

Subject – Techno-Commercial Offer of Linear Arm Sorter

Dear Ajinkya,

Thank you very much for requesting Falcon Autotech for a **sortation system** in your warehouse in **Hyderabad, India**. We are pleased to submit our revised Techno-Commercial Offer in response to your requirement.

The system configuration is aligned with DTDC for optimum positioning in the warehouse as well as operational efficiency.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to being your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies and references.

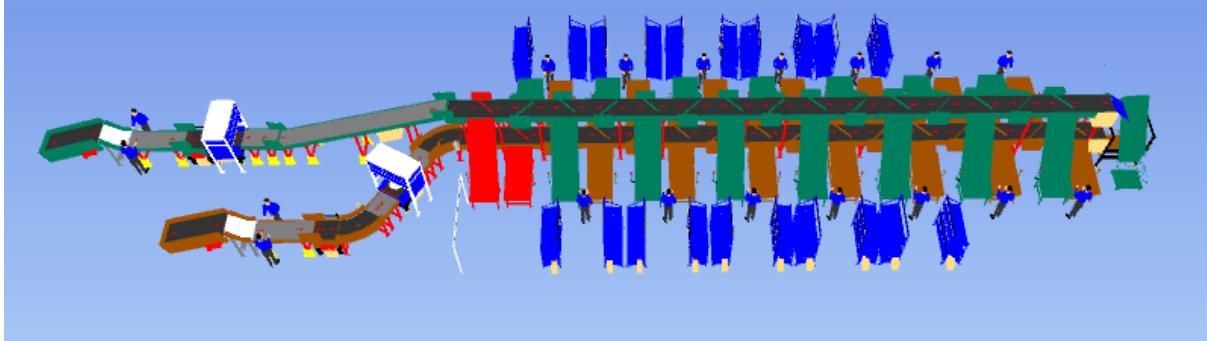
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you with any further information or clarifications that you might have.

Best Regards on behalf of the team,

Rishabh Jain

VP-Sales

Response to RFP for Linear Arm Sorter_DTDC_Hyderabad



Proposal Reference – F25-00170-05

Falcon Autotech Private Limited
Plot No. 87, Sector Ecotech-1, Extension-
1, Greater Noida, Uttar Pradesh 201308.

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1. Glossary

S. No.	Term	Description
1	RFP	Request For Proposal
2	PPH	Parcels Per Hour
3	ICR	Intelligent Character Recognition
4	AC	Alternating Current
5	DC	Direct Current
6	PLC	Programmable Logic Controller
7	IT	Information Technology
8	BOQ	Bill Of Quantity
9	I/O	Input/ Output
10	PDP	Power Distribution Panel
11	PC	Personal Computer
12	UPS	Uninterrupted Power Supply
13	LINEAR ARM	Swivel Wheel Diverter
14	Nos	Number / Pieces
15	DAP	Design Approval Process

2. Executive Summary

Falcon is pleased to confirm its great interest in responding to this DTDC Requirement of Linear Arm Sorter. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening and understanding your needs and ensuring this project's success.

Following the same objective for the Iran system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.

Our solution is based on the following key characteristics:

- **Double Deck Linear Arm sorter offers a designed throughput of 6,000 pph.**
- **14 Nos Primary Sorting Location on each deck**
- **240 Nos PTL Location**
- **1 Rejection Chute on each Deck**
- **1 Sort Fail Chute on each deck**
- **5 Nos PTL on Rejection Chute**

A tailor-made and simple layout designed explicitly for DTDC. The proposed layout is the result of the technical requirements in the RFP document and our discussions with the relevant stakeholders during our meeting.

1. Falcon's reliable Parcel sortation systems

These systems are globally being used by most innovative brands such as Amazon, Asendia, Fastway, Delhivery, Flipkart and many more. The primary and critical components of the FALCON Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from some of the best suppliers in the world, such as SEW, Siemens, Omron, SICK, Faigle, Vahle and Forbo. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

2. Commitment to quality systems

Demonstrating Falcon's clear commitment to the DTDC Group's satisfaction, the parcel sortation system, parts and services will be under warranty for 12 months from installation go-live.

3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

Falcon 2.0

Operational since in 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces

Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS

Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

5 Product Lines

15+ Countries with Live Installations

1800+ Total Installed Systems Globally

1Million+ Sorts Per Hour

600+ Employees

Long-standing Relationships with Industry leaders

News & Articles

FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING
 Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.

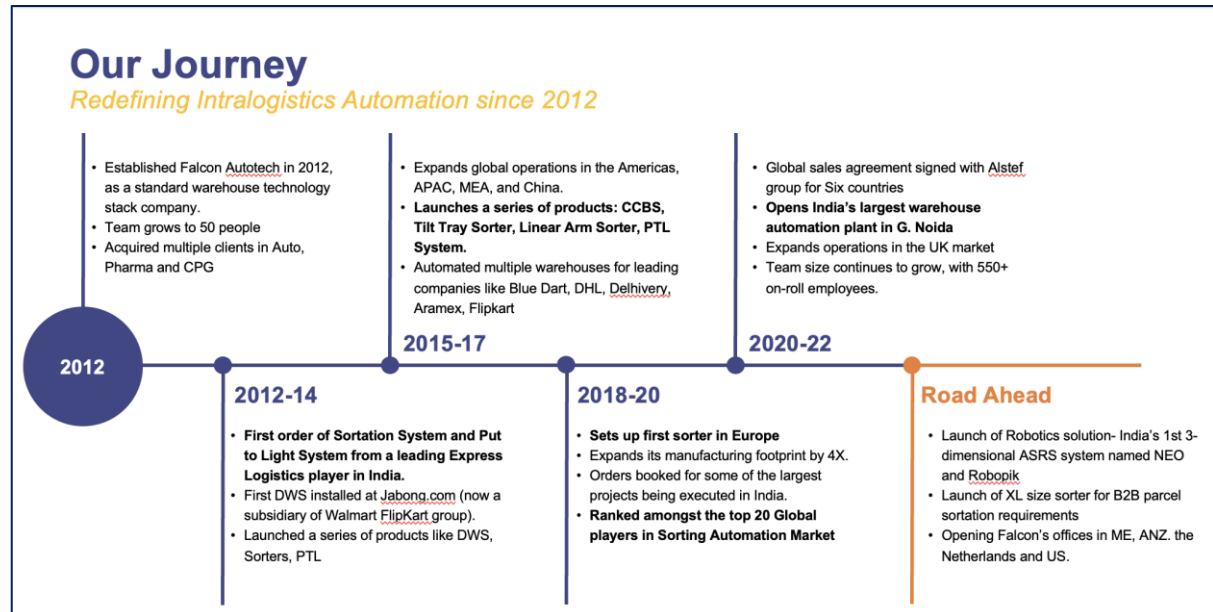
FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP
 Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions
 Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation company, our vision is to become top 10 intralogistics automation company in our focused product lines.

Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.



Product and Solutions

With 100% focused on Parcels, Eatches, Totes, Bags, Cartons



SORTATION SOLUTIONS

- Cross Belt Sorter
- Linear Arm Sorter
- Swivel Divert Sorter
- Tilt Tray Sorter
- Popup Sorter
- Sweep Sorter
- Pusher Sorter



PICK/PUT TO LIGHT SYSTEMS

- PTL Module
- Racks
- Conveyors
- Hand Scanners
- Printers
- Peripheral Displays



DIMENSIONS & WEIGHT SCANNING SYSTEMS

- Cubizon Series
 - R, R-Eco, R-Thru, R-Cross
- Dynamic Profilers
 - Mini (600MM)
 - Jumbo (1200MM)



CONVEYOR SOLUTIONS

- Belt Conveyors
- Roller Conveyors
- Modular Conveyors
- Special Application Solutions



ROBOTICS

- NEO
- Robopick

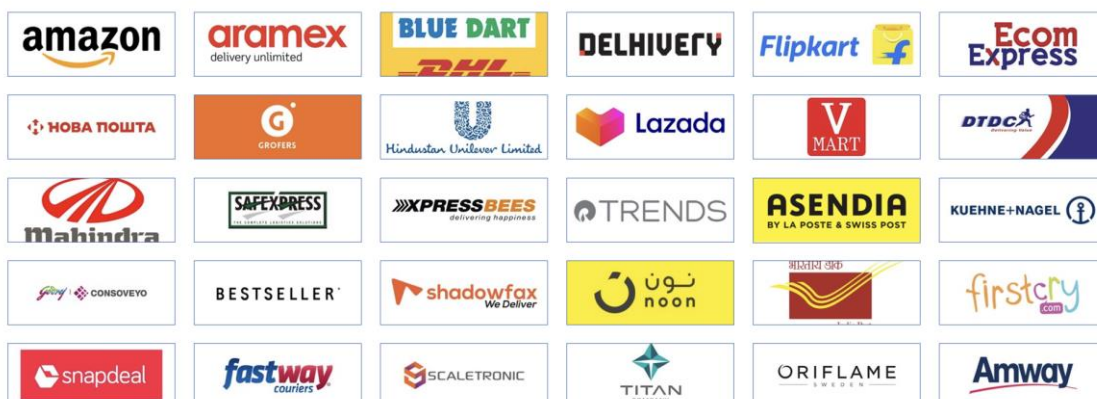


Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements

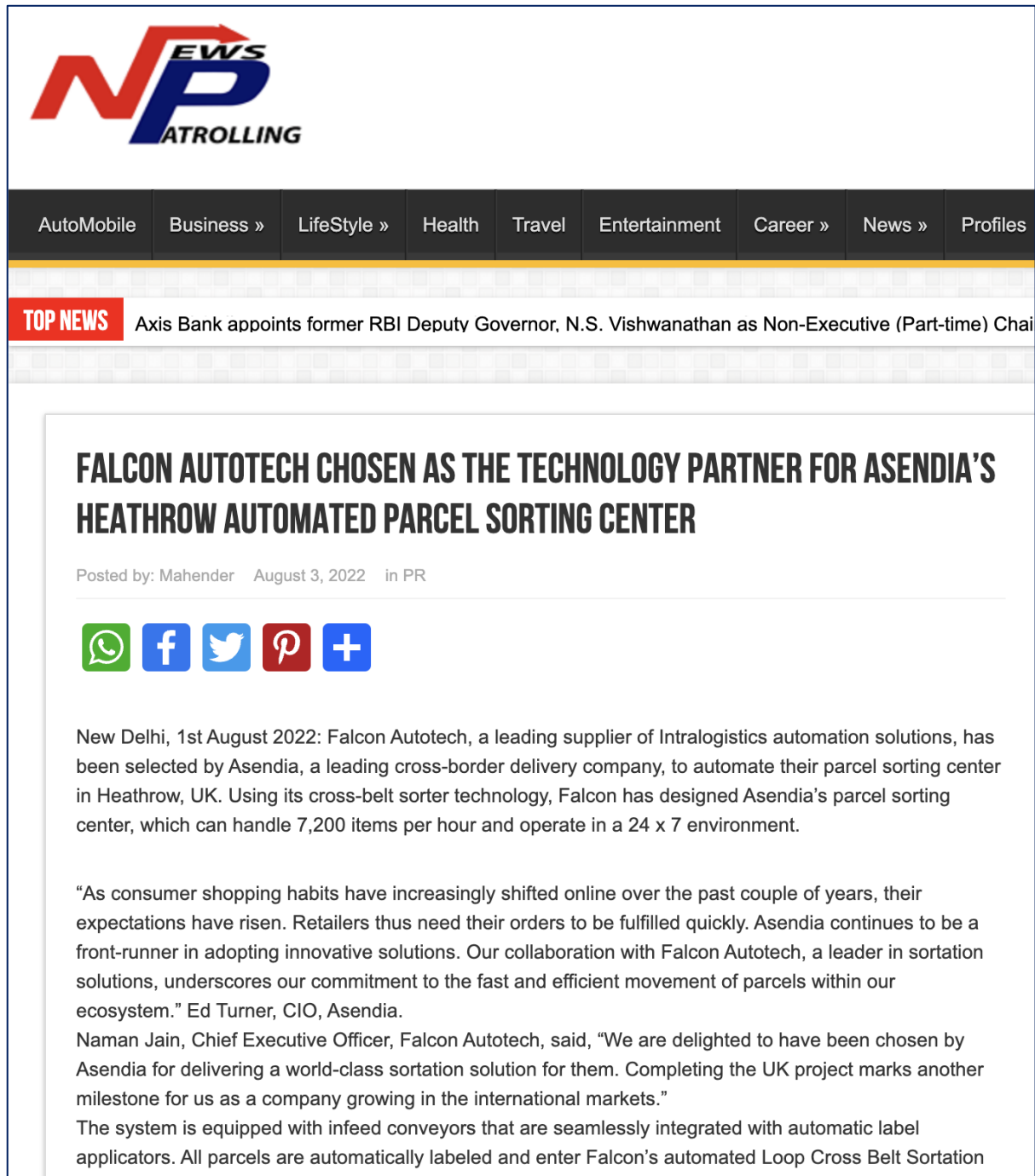


With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.



4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Parcel Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

5.1 Project 1- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 parcels per hour. The system is equipped with three infeed conveyors integrated with automatic label applicator before parcels enter the sortation system. The parcels are sorted using Falcon's Linear Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Linear Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.2 Project 2- (CEP Client, Iran)

In 2019, customer selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing of parcels arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Parcel handling.
- FOCR Engine.
- WCS Software System.

Site Picture –



5.3 Project 3- (Client – Courier Express Parcel, Dubai & Jeddah)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 parcels per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities to into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

Solution Specifications –

- Throughput: 3000 PPH
- End Destinations: 480 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

Site Picture –



5.4 Project 4- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 parcels per hour with the help of Falcon's Linear Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the parcels at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight parcels.

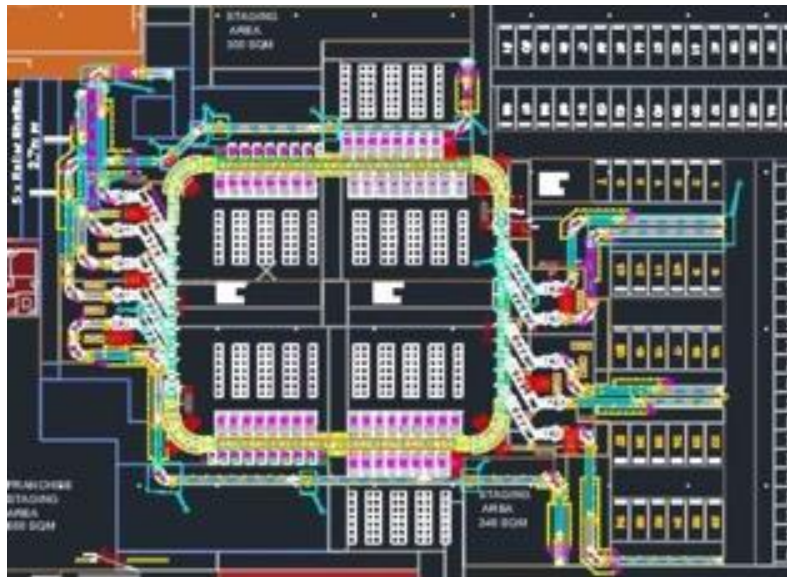
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize parcel.
- Linear Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.5 Project 5- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Parcels.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C parcels. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

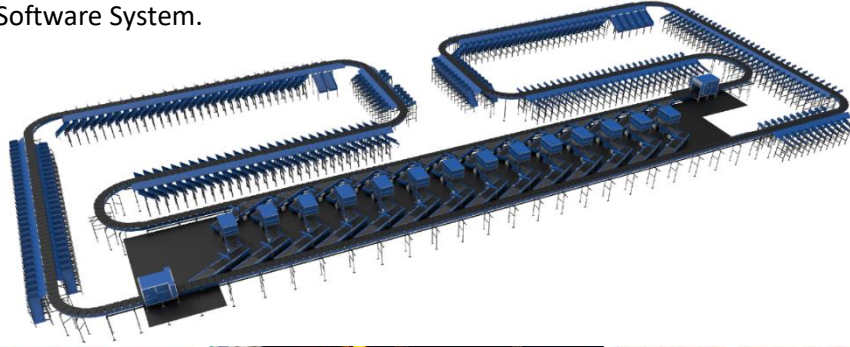
Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.

- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Linear Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.



5.6 Project 6- (Ecommerce Client, India)

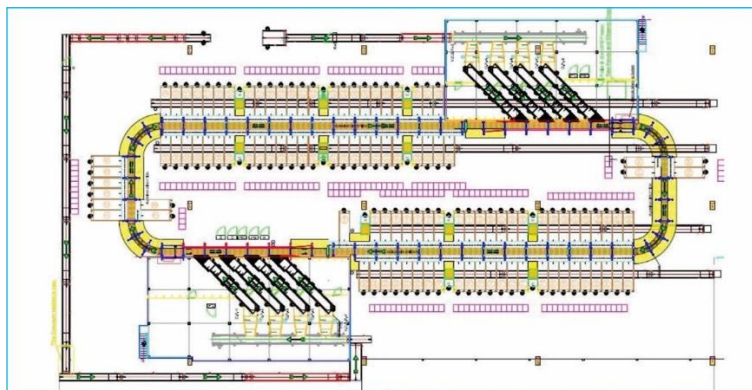
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Linear Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



6. Handled Parcel Spectrum

As per the parcel data provided by DTDC, Falcon has studied and analysed the parcel spectrum in detailed.

Falcon purposes to use its "LINEAR ARM Sorter" to provide the maximum benefits to DTDC in-terms of handling parcel sizes and weight.

6.1 Parcel size loadable on the sorter

Falcon's Linear Arm sorter has a capability to handle the below mentioned parcel sizes and weight.

Specification	Unit	Value
Max Length	mm	600
Max Width	mm	600
Max Height	mm	600
Max Weight	Kg	15
Min length	mm	100
Min Width	mm	50
Min Height	mm	5
Min Weight	gm	100
Dimension Accuracy	mm	+ - 10
Weighing Accuracy	gm	+ - 60 gms or 0.5% whichever is higher

Note- Dimension will only be captured for parcels > 20 mm in height.

6.2 Parcels to be loaded on Sorter shall have the following characteristics:

1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behavior of parcel cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non-conveyable items.
4. Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

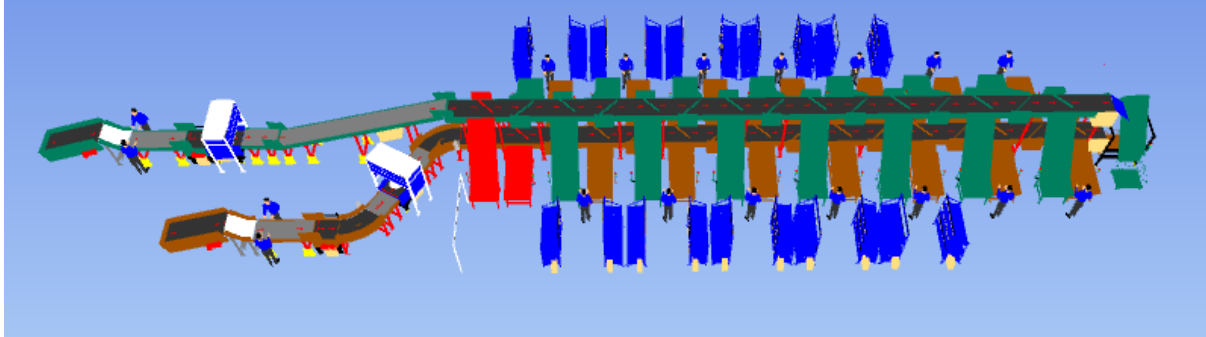
6.3 Parcel not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items.
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g., Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured.
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non-conveyable items.

7. Proposed System Description

7.1 Objective



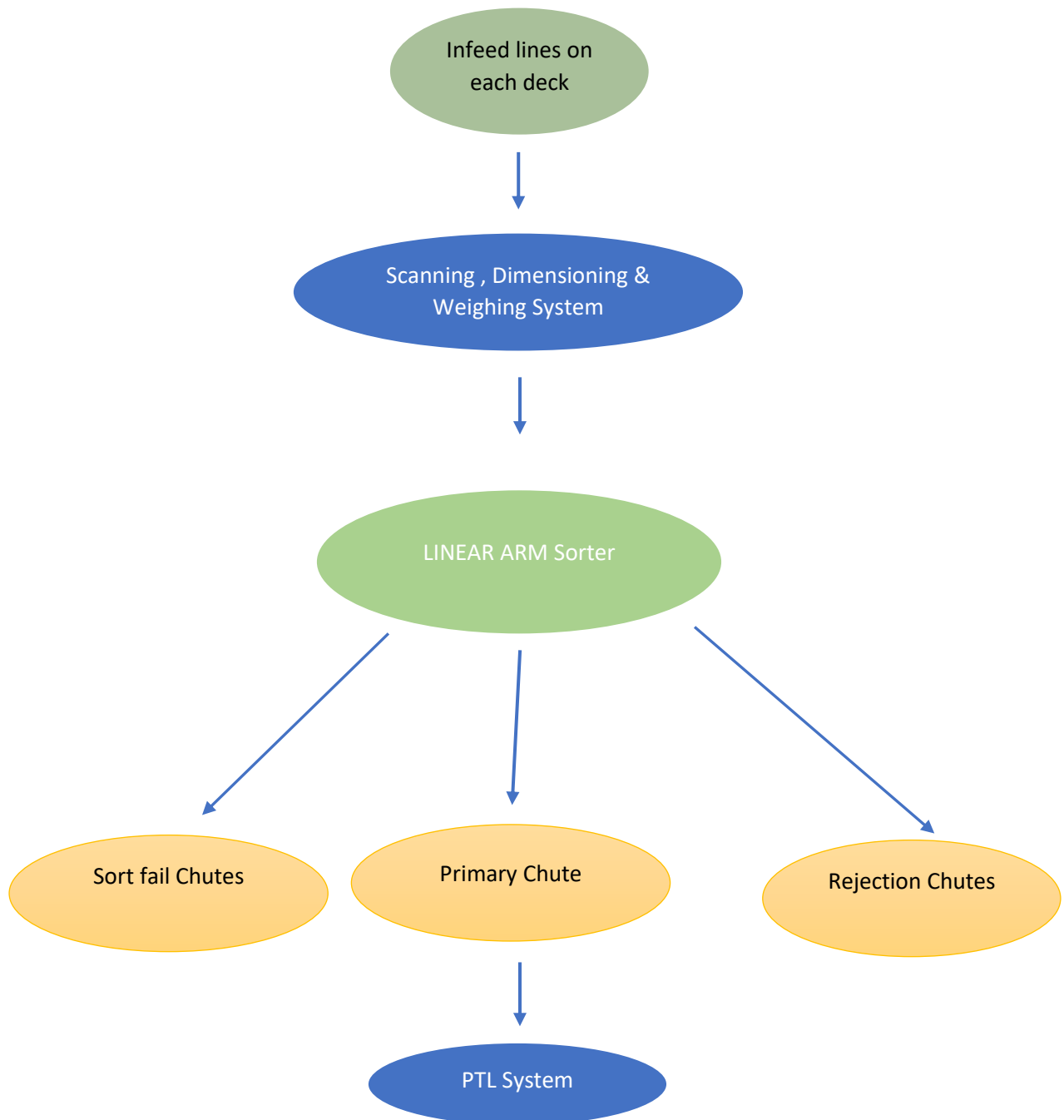
The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing and acceptance testing of the **LINEAR ARM Sorter** for sorting parcels.

7.2 Summary of the System

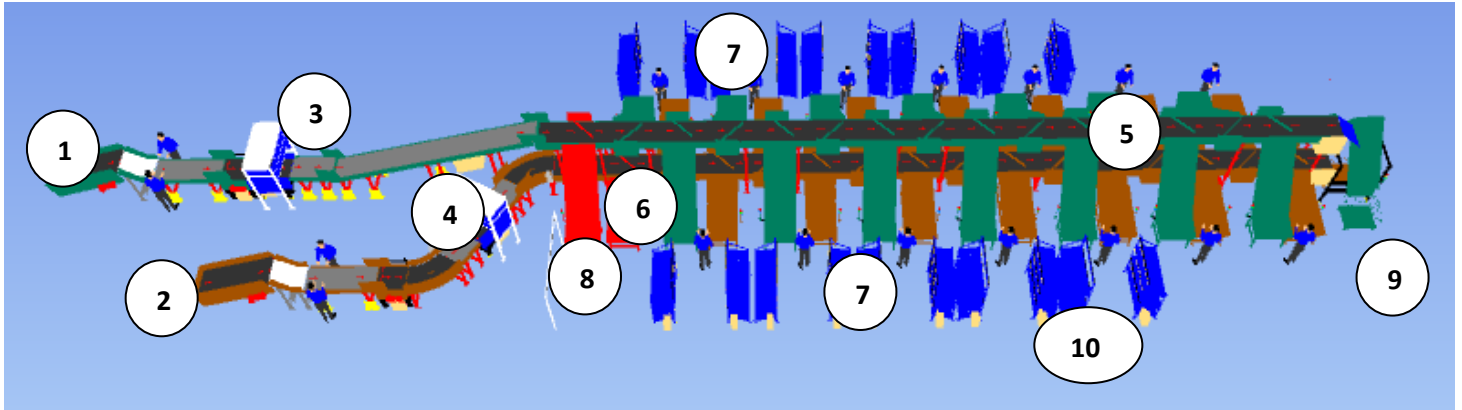
Process flow of the LINEAR ARM Sorter System-

1. **Infeed System:** - Parcel are loaded manually on the loading conveyor present at the beginning of the sorter. Parcel loaded here travel further to ICR for scanning.
2. **Parcel Scanning & Weighing:** - Once the parcels reach the Scanning zone, Set of Buffer conveyors automatically maintain sufficient space between two parcels, Now the singulated parcel will get scanned via scanner & its weight will be captured with the help of dynamic weighing machine.
3. **LINEAR ARM Sorter:** - Once the parcels are on main sorter line, the LINEAR ARM diverters sorts the parcels into respective chutes using data from DTDC Sorting logic.
4. **Output Chutes:** - The shipments are discharged into different types of chutes-
 - a. **Primary- 14** Output chutes (10 No with PTL & 4 Direct Chutes) are present in the system. Parcels sorted in these chutes gets collected in the spoon type chute for secondary sorting and other process.
 - b. **PTL** - On 10 Nos Chute there will PTL i.e. 24 PTL on each chute for doing the secondary sorting
 - c. **Rejection Chute-** One rejection chute is present on each deck to handle rejected parcels.
 - d. **Sort Fail Chute** - One sort fail chute is present on each deck to handle rejected parcels.

8. Concept Description



9. Layout View of LINEAR ARM Sorter



Legend

1. Bulk Loading Conveyor for Upper Deck
2. Bulk Loading for Lower Deck
3. Scanning, Weighing & Dimensioning System for Upper Deck
4. Scanning, Weighing & Dimensioning System for Lower Deck
5. Linear Arm Sorter Upper Deck
6. Linear Arm Sorter Lower Deck
7. Primary Chute
8. Rejection Chute
9. Sort Fail Chute
10. PTL Rack

10. Proposed System Capacity Calculations

10.1 Sorter System Capacity

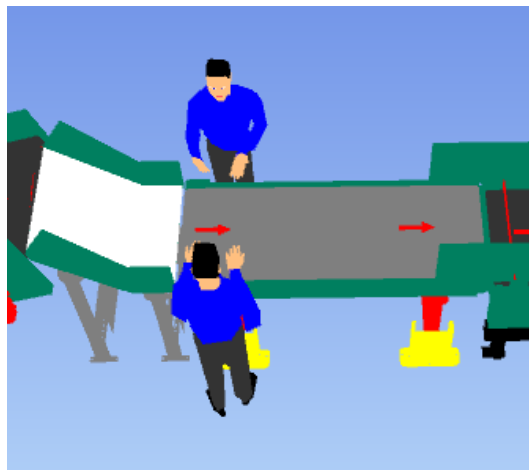
The following table shows the throughput calculation for the sortation system designed based on DTDC RFP requirements.

SPECIFICATION	VALUE
Sorter Type	LINEAR ARM Sorter
Designed TPH	3000* PPH on each deck
No of Loading point	1 Nos on each deck

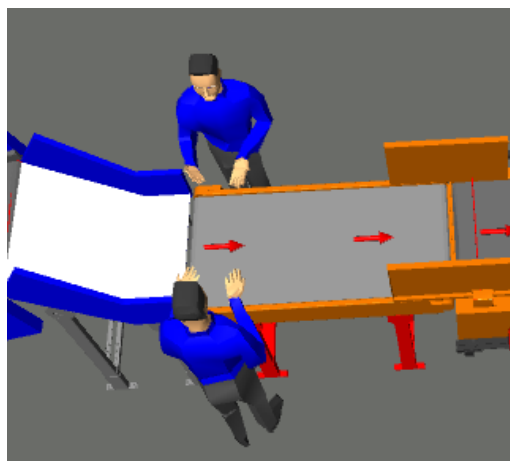
Note : This is designed TPH on Avg parcel size. Operational TPH will be based on the feeding capacity of the operators.

11. System description

11.1 Infeed System



For Upper Deck



For Lower Deck

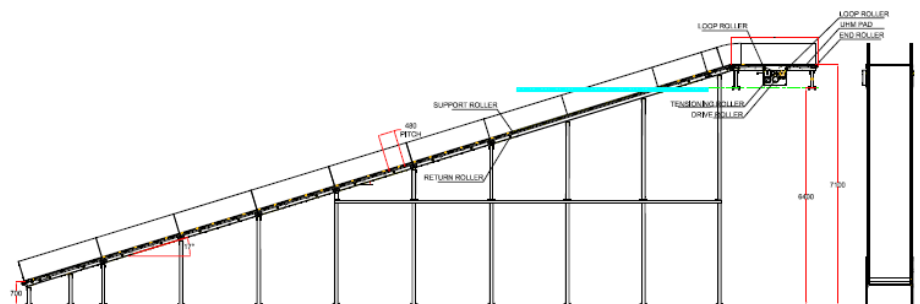
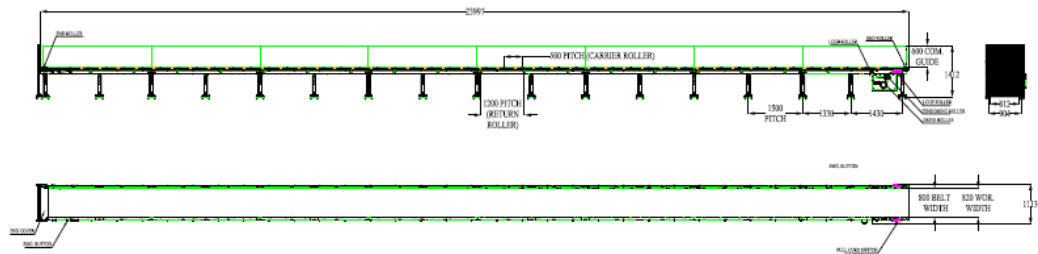
11.1.1 Straight and Inclined Powered Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.

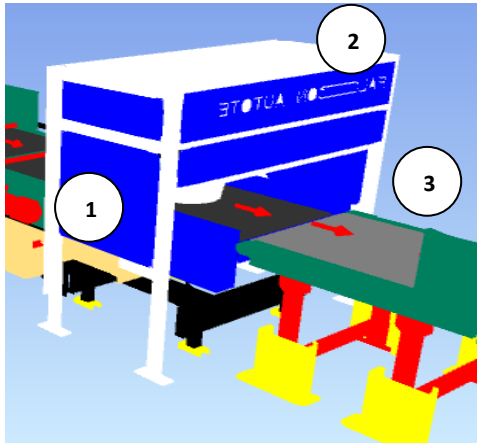
Some Silent Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.

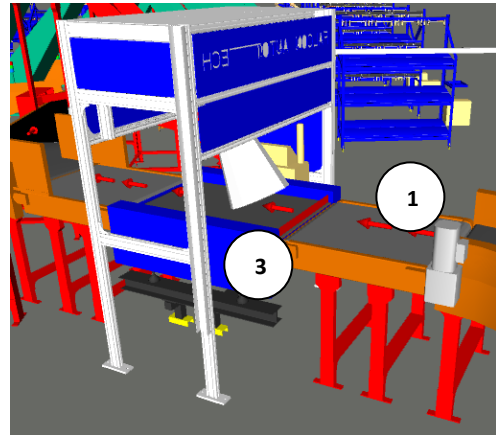


Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	800
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Motor Shaft	MOC	EN9 Shaft material
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Load capacity per unit length of Conveyor	kg/m	50
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord/Lenze
Drive	make	Lenze/Omron/Allen Bradley/Similar
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

11.2 Automatic Scanning & Weighing



For Upper Deck



For Lower Deck

Legend

- 1- Spacing Conveyor
- 2- Scanning Tunnel
- 3- Weighing Conveyor

11.2.1 Automatic Barcode Scanning & Dimensioning System

System is Integrated with Automated Line Scan Camera Technology for scanning 1D & 2D Barcodes

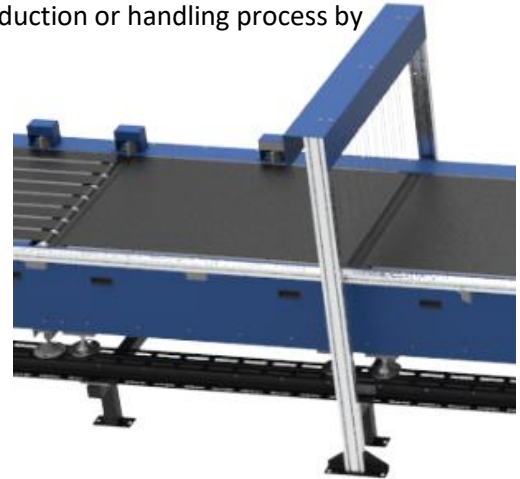
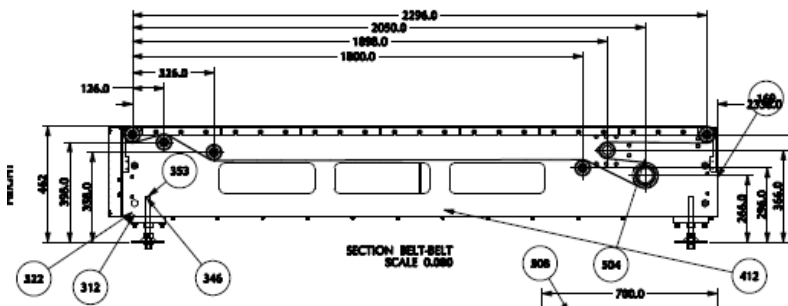
Specification	UOM	Remark
Manufacturer Name	Name	Cognex
Barcode Type	Type	1D & 2D
Module Size	Mils	≥ 10 for 1 D & ≥ 25 for 2D
Shipment Size	Mm	Min: 100 x 50 x 5 Max: 600 x 600 x 600
Number of Codes	Nos	1
Location	Side	Top
Orientation	Type	Top side
Colour of bars	Colour	Black
VMS	Nos	1
Under Foil	yes/no	No



11.2.2 Spacing Conveyor

Spacing conveyor is used to temporarily store or hold parcels in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

Buffer conveyor is particularly useful in scenarios where there is a discontinuity or variation in the flow rate of items. They allow for a smooth and continuous production or handling process by absorbing temporary surges or gaps in the flow.



11.2.3 Weighing Conveyor

A weighing conveyor, also known as a weigh belt conveyor, is a type of conveyor system specifically designed to measure the weight of materials as they move along the conveyor belt. It combines the functions of conveying and weighing into a single integrated process.

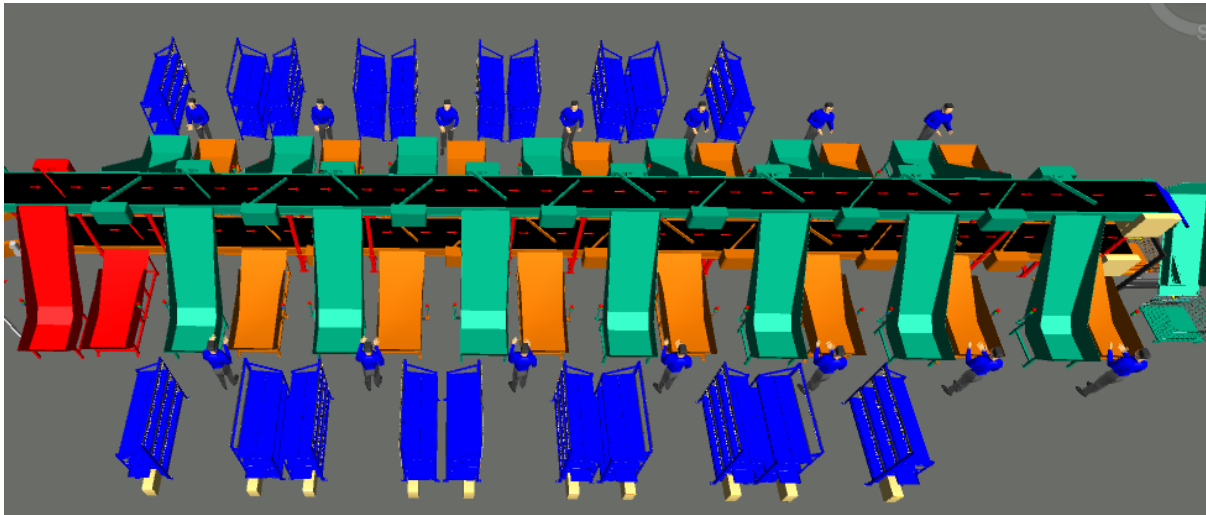
Weighing Conveyors equipped with high precision Load Cells to capture the weight of Parcels.

Make- Bizerba/ Mettler/Equivalent



11.3 LINEAR ARM Sorter

LINEAR ARM is a non-impact sortation module for linear sortation lines, ideal for medium volume sort centres. LINEAR ARM's modular design allows it to save space while also allowing for future expansion of sorting point

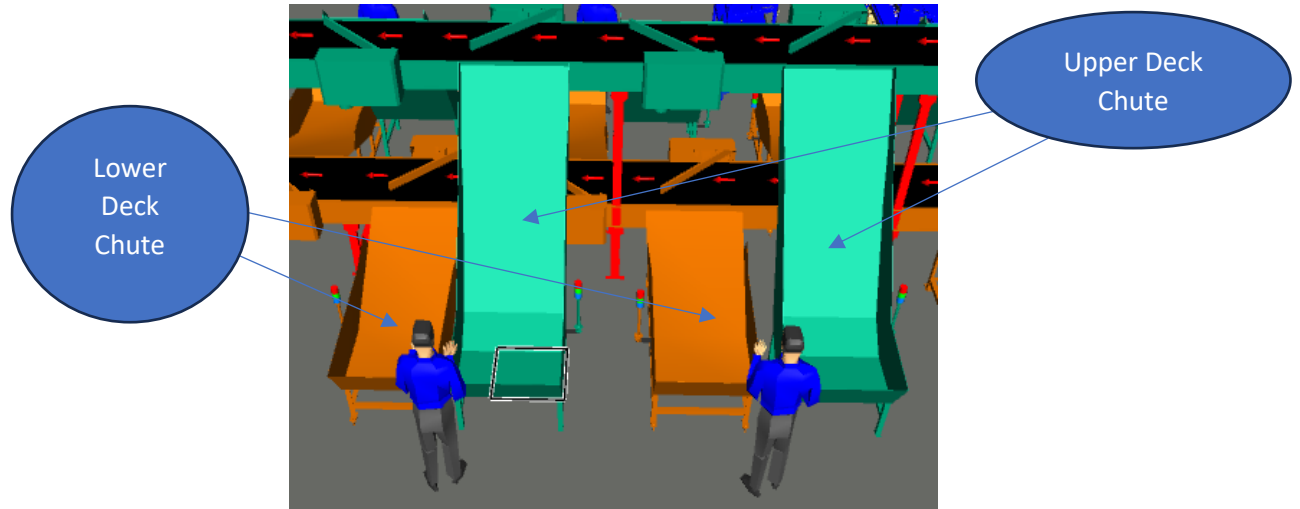


1. The proposed Double Decker LINEAR ARM Sorter will be installed on ground level. The upper deck runs at 2700 mm from the ground level and lower deck 1000 mm from the ground level

2. As Parcels pass through the Barcode Scanning Tunnel on the LINEAR ARM Sorter, their barcodes are scanned, and chutes are assigned. Once the shipment reaches the respective chute, the LINEAR ARM Diverter unit diverts the shipment and drops the shipment in the assigned chute.

11.3.1 Primary/Collection Chute

In the proposed solution, 14 Nos primary chute has been planned on each deck .

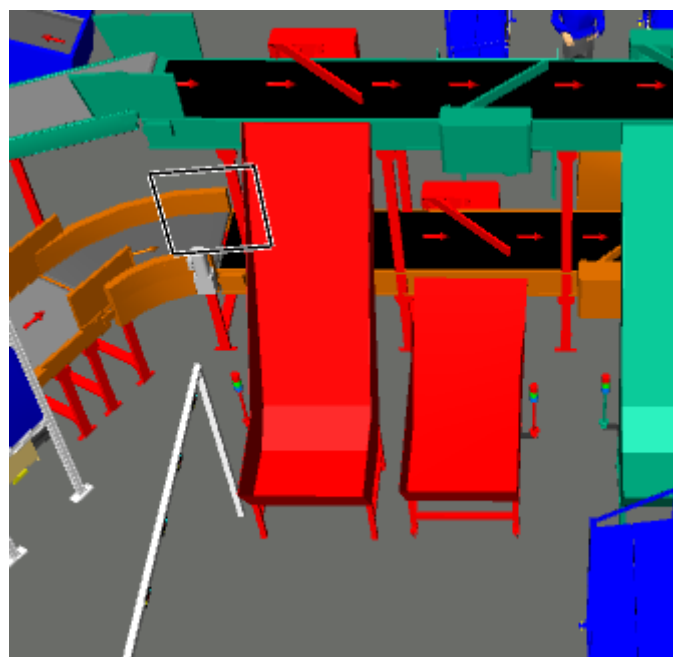


On each chute below accessories are considered:

- Chute Full Sensor: 1 No
- Tower light: 1 No

11.3.2 Rejection Chute

In the proposed solution, 1 No chute has been planned on each deck .



11.3.3 PTL Light & Racks

PTL Racks & Modules

- PTL Racks with PTL light modules has been provided for secondary sorting of parcels that have been dropped in the PTL chutes.
- Rack Configuration: 3 Rows, 4 Columns, total of 12 modules * 2 racks in one chute.



Process

- Operators will pick up a parcel from the PTL /Collection chute and scan the parcel barcode by handheld scanner.
- The light of the block in the PTL rack which has been mapped to the destination location of the parcel will glow.
- Operator will place the parcel in the bag framed in that module and acknowledge by pressing the confirmation button on the module.
- This process will be repeated for all parcels falling in that chute bin.
- Once a bag is full, the operator will take out the bag, paste specific tag on bag seal and remove bag from the secondary sortation area.
- The new bag will be placed in the empty module.

12. Proposed System Technical Details of Linear Arm Sorter

12.1 Mechanical equipment

For Upper Deck			
Pos.	Qty.	Description	Value
1	1	Infeed System - Powered Belt Conveyors - Spacing Conveyor - Curve Conveyor - Launch Pad Conveyor with Chute - Barcode Scanner - Dimensioning System - Weighing Conveyor - Hand scanner for Manual Mode	~10 metres (3 Conveyors) 1 No 1 No 1 No 1 No 1 No 1 No 1 No
2	1	LINEAR ARM Sorter 1 LINEAR ARM Sorter - Sorter Height Including: - LINEAR ARM Diverters	2700mm (from ground level) 15 Nos
3	1	Sorter Outputs - Primary/Collection Chute - Rejection Chute - Sort Fail Chute - Fixed Bin - Chute Full sensor - Tower light (2 Colour)	14 Nos 1 No 1 No 1 No 16 Nos 16 Nos
4	1	PTL Racks - PTL Light - PTL Rack 1 D + 2D Hand scanner	240 Nos 240 Nos 15 Nos
5	1	PTL Frame for Rejection Chute - PTL Light - PTL Frame	5 Nos 1 No

For Lower Deck			
Pos.	Qty.	Description	Value
1	1	Infeed System - Powered Belt Conveyors - Spacing Conveyor - Curve Conveyor - Launch Pad Conveyor - Barcode Scanner - Dimensioning System - Weighing Conveyor - Hand scanner for Manual Mode	~3 metres (3 Conveyors) 2 Nos 2 Nos 1 No 1 No 1 No 1 No 1 No
2	1	LINEAR ARM Sorter 1 LINEAR ARM Sorter - Sorter Height Including: - LINEAR ARM Diverters	1000mm (from ground level) 15 Nos
3	1	Sorter Outputs - Primary/Collection Chute - Rejection Chute - Sort Fail Chute - Fixed Bin - Chute Full sensor - Tower light (2 Colour)	14 Nos 1 No 1 No 1 No 16 Nos 16 Nos

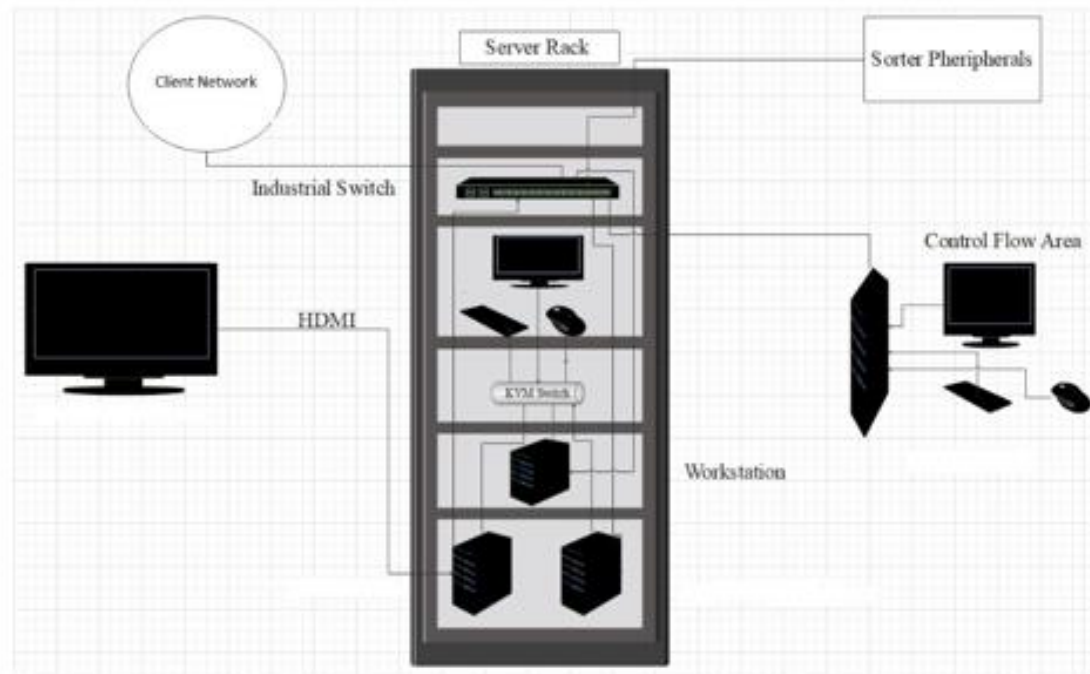
12.2 Electrical Equipment

Electricals			
1	1	Consists of Main Power Distribution panel Main Control Panel Network Switches Field Cabling	Included

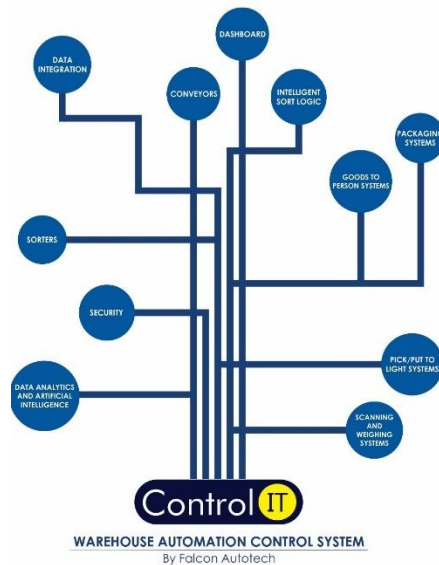
12.3 IT Equipment

IT			
1	1	PC Server	2 Set

13. IT Architecture (For Reference Only)



14.Falcon's WCS CONTROLIT (For Reference Only)



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

15. Key Components Make

Items	Make
Belts	Forbo
Rollers	Falcon
Weighing Scales	Bizerba
Barcode Scanner	Cognex
Volumetric Scanner	Cognex
Sensors	Sick/Leuze/ P&F
PLC	Omron
Control Panels	Rittal/ BCH
VFDs	Lenze/ AB/ Omron/Similar
Motor	SEW/Nord
Cables	LAPP
Switch Gear	Schneider
Bearings	NTN/SKF/Excluding Bearing of Rollers. Bearings will be FAPL /OEM standard)
HMI's	Omron
LINEAR ARM	FALCON

16.DTDC Responsibility

16.1 DTDC Responsibilities During the Assembly and Commissioning Phase

- Provision of the site complex and office area facilities.
- The possibility of authorizing access to the site and the execution of the installation work up to 6 days a week and 24 hours a day if deemed necessary and if requested by FALCON.
- Free provision, during the installation phase, of the power supply necessary for the installation activities (estimated at 20 kW).
- Provision, during the commissioning phase, of the power supply necessary for the operation of the parcelsorting system free of charge at the date of FALCON need.
- Provision of the IT system functionality in accordance with the specification at the date of FALCON need.
- The customer is responsible for a safe working environment.
- The customer makes arrangements for the working area('s) to be protected against direct weather influences.
- The customer provides adequate lighting, heating, and ventilation to create a normal working environment.
- The cost of temporary storage that may be required (other than in the immediate vicinity of the installationsite) is not included in the scope of delivery of this quotation. This also applies to temporary storage that may be required because materials are ready for delivery (in accordance with the schedule) but cannot be delivered due to hold-ups on the customer's side.
- The customer is responsible for the demarcation of aisles and danger zones with floor paint.

16.2 Responsibilities of DTDC During the Tests

- Provision of the test loads and barcode labels required for the tests.
- Provision of personnel required for test activities (loading and unloading operations).
- Provision of the necessary information to sort the parcels correctly.
- Verify with FALCON the quality and conformity of the test loads (labels, cartons).
- Will need to provide the necessary staff to collect information on the tests and to verify and confirm testresults with FALCON.

16.3 DTDC Responsibilities During the Training

- Free from their usual work, the employees participate in the training for the duration of the training.
- Provision of a list of participants for each available training course 3 days before the start of the course.
- Provision of a classroom equipped with a whiteboard, video projector, projection

screen, and enough space for desks or tables and chairs for the trainer and trained staff.

- Check the prerequisites of the people who have to follow the training, e.g. the qualifications of the technical staff.
- The invitation of staff to attend the training courses will be at the expense of DTDC.

17. Commercial Terms

17.1 Price Sheet

Sr. No	Component	Qty per set	Set	UOM	Prices
1	Double Deck Arm Sorter Package	1	1	Set	₹ 2,44,00,000

17.2 Price Breakup

Sr. No	Component	Qty per set	Set	UOM	Prices
1	Double Deck Arm Sorter Package	1	1	Set	₹ 1,26,81,677
2	Conveyors Package	1	1	Set	₹ 38,84,943
3	Chute package	1	1	Set	₹ 28,69,044
4	PTL Package	1	1	Set	₹ 21,95,800
6	Safety & Software Package	1	1	Set	₹ 9,10,154
7	Installation & Commissioning	1	1	Set	₹ 11,07,081
8	Packaging & Documentation	1	1	Set	₹ 4,42,832
9	Project Management and Engineering Charges	1	1	Set	₹ 4,81,831
10	Transportation & Unloading Charges	8	1	Trucks	₹ 10,00,000
11	3 Months Operational Support (1 Engineer)	1	1	Set	₹ 4,50,000
Total package					₹ 2,60,23,361
Strategic Discount					₹ 16,23,361
Final Price					₹ 2,44,00,000
Critical Spare					₹ 17,00,000

CMC Package

Year	2nd Year	3rd Year	4th Year	5th Year	6th Year	7th Year	8th Year	9th Year	10th Year
CMC Cost	₹ 10,55,723	₹ 10,55,723	₹ 10,55,723	₹ 11,61,295	₹ 12,66,868	₹ 13,72,440	₹ 14,78,012	₹ 15,83,585	₹ 16,89,157

The Total Price is to be understood.

- Freight – Included
- Taxes as applicable - Excluded
- Packing and Forwarding - Included
- Server: Normal PC Type 2 Nos Included for both deck
- Material Unloading at Site & Transit Insurance: Included
- Operational Support: 1 Engineer for 1 Shift upto 3 Month post go live.
- HSN Code : 84798999

17.3 Payment Terms

Payment Percentage	Stage
40%	Advance along with PO
50% + 100% Taxes	Against material delivery at site
10%	After Installation & Commissioning

**All invoices (if due as per payment terms) should be cleared within 7 days from the date of receipt of undisputed invoice by the buyer*

17.4 Delivery Timelines

Go Live Date of 31st December 2025 considering the PO receipt date on or before 15th Aug 2025 .

18. System Life

This system is majorly a mechanical component if well maintained as per OEM specification, Falcon can provide system support for lifetime. The life of system will be minimum 10 years and even up to 15 years.

Please note this is fully dependent on system health status which will be carried out by Falcon at regular intervals.

19. Warranty Period

Falcons offered System comes with a standard warranty of 1 year (Starts from the date of beneficiary, max 30 days after readiness of commissioning) and post completion of warranty, customer can opt for an Extended warranty. The warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Hotline Support when required
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in the Exclusion Clause)

The warranty does not apply to the replacement or repair of:

- Normal wear and tears.
- Consumables. (List can be shared during DAP)
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that is supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

Note: Transportation cost for sending faulty items to Falcon Factory from the site will be in Client Scope and new items from Falcon factory to site will be in scope of Falcon.

20.Comprehensive Maintenance Contract

20.1 Scope of Service

The following services would be covered under this contract:

- E-mail support, Telephonic support, Remote assistance via Team Viewer will be extended to the customer for all issues.
- Preventive Maintenance of the System as per FAPL schedule.
- On-site breakdown service visits & Regular software updates.
- Software related small change requests such as change in the report format, User Interface related changes such as font size, specific color etc.
- Supply, repair/replacement of faulty spare*.
- Supply, replacement of consumables.
- One day workshop for operator training will be provided twice during this service contract.

Sr. No.	Services	Comprehensive Annual Maintenance Package (CMC)
1	Remote Support	✓
2	Break Down Service Visit	✓
3	Preventive Maintenance Visit	✓
4	Software Updates	✓
5	Small Change Request	✓
6	Consumables	✓
7	Spares Warranty	✓
8	Operator Training	2 x one day workshop for operators at client site

*Replacement of spare is subject to failure analysis and approval by FAPL.

20.2 Requesting Service

- Email: support@falconautoonline.com
- Support Portal: <https://falconautotech-pvt-ltd.my.site.com/>
- Working Hours: 24 x 7

20.3 Service Response

The following will be the service response for all issues reported to FAPL.

1. Telephonic and Remote support

Response Time	Severity-0: 60 min Severity-1: 60 min Severity-2: 120 min Severity-3: 120 min
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2. On site physical engineering visit, in case if required.

Engineer Visit Tier-1 City	Severity-0: 24 Hour Severity-1: 24 Hour Severity-2: 36 Hour Severity-3: 48 Hour
Engineer Visit Tier-2 City	Severity-0: 36 Hour Severity-1: 48 Hour Severity-2: 48 Hour Severity-3: 72 Hour

3. Some Examples of Severity

Severity 0 - System Down	1. Entire loop is down 2. All feedlines are down 3. Server down
Severity 1 - Critical Impact	1. One Zone operation completely down 2. Multiple feedlines affected 3. VDS completely down
Severity 2 - Moderate Impact	1. Any one Feedline affected 2. Bagging conveyor belt damage 3. Any infeed conveyor down

20.4 Service Level Agreement (SLA) for Spares and Consumables dispatch

Spares & Consumables dispatch	Severity-0: 24 Hour * Severity-1: 24 Hour * Severity-2: 48 Hour * Severity-3: 48 Hour * *If stock available in Plant (Monday 9:00 AM to Saturday 6:00 PM) Excluding Public Holidays
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***If stock is not available in plant OEM lead time will be followed.*

Spares and Consumables transportation responsibility

- The transportation cost of spares & consumables will be in **FAPL scope**.
- The transportation cost of faulty/damaged spares back to Falcon will be in **customer scope**.

20.5 Escalation matrix

Escalation Matrix			
Level	Name	Email	Contacts
1	Nitin – South Zone	nitin.Srivastava@falconautoonline.com	7812997692
2	Sandeep Sharma for sorters	sandeep@falconautoonline.com	85888 80764
	Manish Kumar for non-sorter products	manish.kumar@falconautoonline.com	8447731157
3	Manuj Bansal	manuj@falconautoonline.com	99100 85558
4	Ankit Jain	ankit.jain@falconautoonline.com	81300 65528

20.6 General Terms and conditions with Customer Responsibilities

1. The system entering into the contract should be in good working condition.
2. A health check-up of the system will be done by FAPL on a paid basis before entering into the contract.
3. Proper reporting of the Issue faced via the contact routes specified.
 - b. For this it is recommended that customer should train and authorize their employee(s) to coordinate with FAPL (SELLER)
 - c. Trained customer employee(s) shall be available on site during each intervention (remote and on site); this(these) employee(s) shall also apply safety and rescue measures in case of emergency.
4. Data transmission requirements for telephonic and remote support (data stream, internet connection speed etc.) – the customer shall ensure the sufficient quality of the telephone and remote connection; any delays and additional costs caused by insufficient data transmission or lack of cooperation from the on-site personnel shall be borne by the customer.
5. The availability of all parts which require customer-specific programming and configuration, within the agreed reaction times for on-site interventions is not guaranteed; the customer shall ensure the on-site availability of these parts in order to avoid delays for the on-site interventions.
6. To send any faulty/damaged parts with proper packaging and documentation back to FAPL, Greater Noida-India, facility for RCA, whenever required.
7. Parts purchased from FAPL must be used and maintained by the customer.
8. In no event shall SELLER be liable to BUYER for any consequential, indirect, special, or punitive damages, including but not limited to loss of business, goodwill, income, profit or savings as well as any other claim for consequential compensation arising out of or relating to this service contract.
9. The BUYER is not authorized/allowed to replace, adjust, repair or in any way manipulate the automated system and its components unless he is specially trained and authorized to perform the corresponding manipulation by FAPL.

20.7 Exclusions

- New IT development. Training, Data Backup, OS Support, Application Support and Virus Support.
- Any damage to the equipment due factors such as wrong operation, negligence, improper use, physical damage, damage due to electrical fluctuations/surges/ etc., damage due to Fire, Act of God or any other damage that cannot be attributed to the OEM/FAPL. In such cases, FAPL will take all reasonable steps to ascertain the RCA in coordination with the OEM and the final decision of OEM shall be binding on buyer.
- Dismantling & re-installation of machine related activities.
- Modification/upgradation of the existing setup

21. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Pallet boxes / Roll Cage Trolley / Bags
- Construction Power
- Steel Works- If not specified
- SCADA
- Waste Removal from the Installation Area
- Assembly tools like forklifts, ladder and hoisting machines for maintenance
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and network wiring to the main control cabinets
- UPS for Controls and Drives
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.